Ex. No: 3

Title: Probability Distributions and Bayes' Theorem

Packages: NumPy, SciPy, Matplotlib

AIM:

To plot the Normal and Binomial probability distributions and to apply Bayes' theorem.

ALGORITHM:

1. Start the program and import NumPy, SciPy and Matplotlib.
2. Plot the Normal distribution using stats.norm.pdf().
3. Plot the Binomial distribution using stats.binom.pmf().
4. Take P(Disease), P(Positive | Disease) and P(Positive | No Disease).
5. Calculate P(Positive) = P(Pos|D) P(D) + P(Pos|No D) P(No D).
6. Calculate P(Disease | Positive) = P(Pos|D) P(D) / P(Positive).
7. Display the output and stop.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

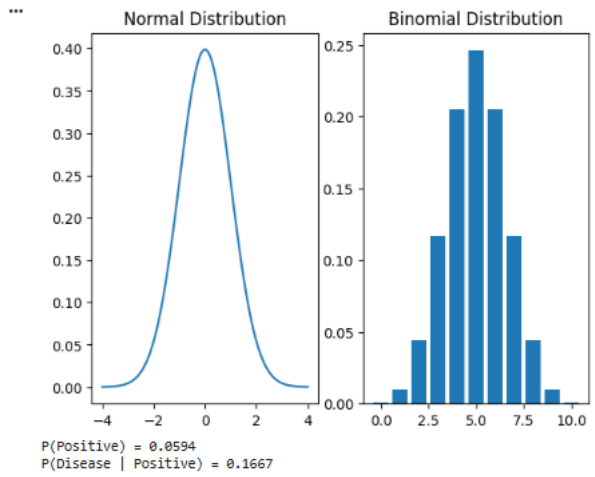
x = np.linspace(-4, 4, 100)
plt.subplot(1, 2, 1)
plt.plot(x, stats.norm.pdf(x, 0, 1))
plt.title("Normal Distribution")

k = np.arange(0, 11)
plt.subplot(1, 2, 2)
plt.bar(k, stats.binom.pmf(k, 10, 0.5))
plt.title("Binomial Distribution")
plt.show()

# Bayes' Theorem
p_d = 0.01          # P(Disease)
p_pos_d = 0.99      # P(Positive | Disease)
p_pos_nd = 0.05     # P(Positive | No Disease)

p_pos = p_pos_d * p_d + p_pos_nd * (1 - p_d)
p_d_pos = p_pos_d * p_d / p_pos
print("P(Positive) =", round(p_pos, 4))
print("P(Disease | Positive) =", round(p_d_pos, 4))
```

OUTPUT:



RESULT:

The probability distributions were plotted and Bayes' theorem was applied successfully. P(Disease | Positive) = 0.1667.